National University of Computer and Emerging Sciences,

Islamabad

FAST School of Computing

CS4031 - Compiler Construction Assignment 01

Automata Design

Submitted by

Muhammad Fatik Bin Imran (23i-0655)

Muhammad Kaleem Akhtar (23i-0524)

Bachelor of Computer Sciences

2025

Lecturer

Dr Mahreen Alam

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1. Regular Expressions

RE (Fate Language)

Keywords:

(start|finish|loop|condition|declare|output|input|function|return|break|continue|else)

1. Identifiers:

[A-Z][a-z0-9_]{0,30}

2. Integer Literals:

[+-]?[0-9]+

3. Floating-Point Literals:

[+-]?[0-9]+\.[0-9]{1,6}([eE][+-]?[0-9]+)?

4. String Literals:

"([^"\n\r\\]|\\["\\ntr])*"

5. Character Literals:

'([^'\n\r\\]|\\['\\ntr])'

6. Boolean Literals:

(true|false)

7. Operators:

Arithmetic: (**|[+\-*/%])

Relational: (==|!=|<=|>=|<|>)

Logical: (&&|\|\||!)

Assignment: (\+=|-=|*=|/=|=)

Inc/Dec: (\+\+|--)

8. Punctuators:

[(){}\[\],;:]

9. Comments:

Single-line: ##[^\n]*

Multi-line: #*([^*]|*+[^*#])**+#

10. Whitespace (skip but track line/column):

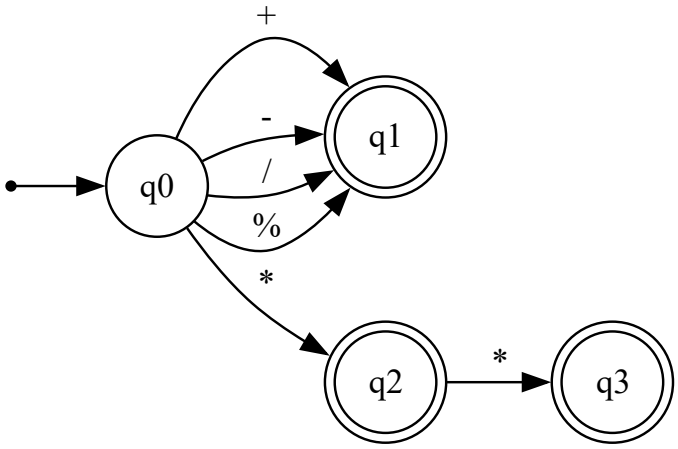
[\t\r\n]+

Priority Order (highest to lowest):

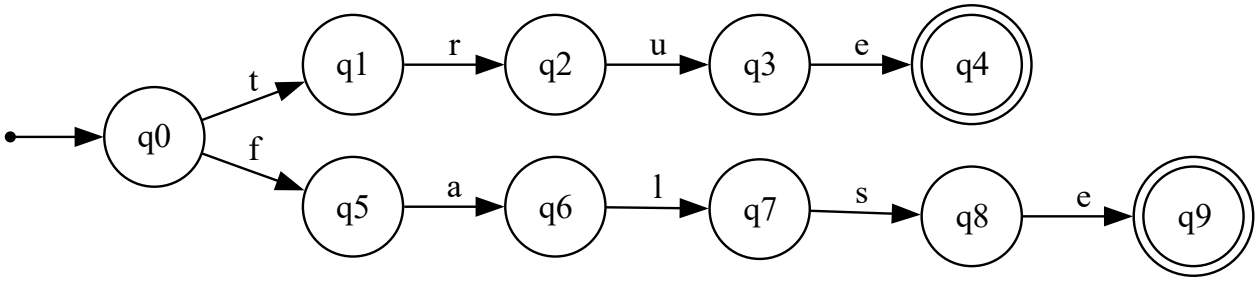
1. Multi-line comments
2. Single-line comments
3. Multi-character operators (**, ==, !=, <=, >=, &&, ||, ++, --, +=, -=, *=, /=)
4. Keywords
5. Boolean literals
6. Identifiers
7. Floating-point literals
8. Integer literals
9. String literals
10. Character literals
11. Single-character operators
12. Punctuators
13. Whitespace

2. NFA

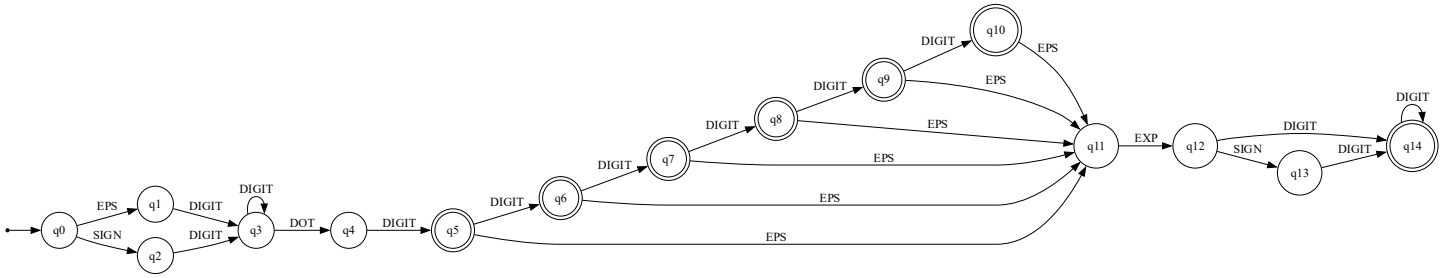
2.1 Arithmetic operators



2.2 Boolean

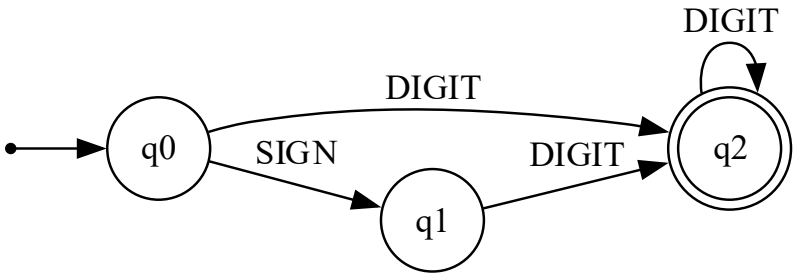


2.3 Float

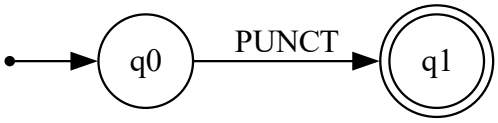


2.4 ID

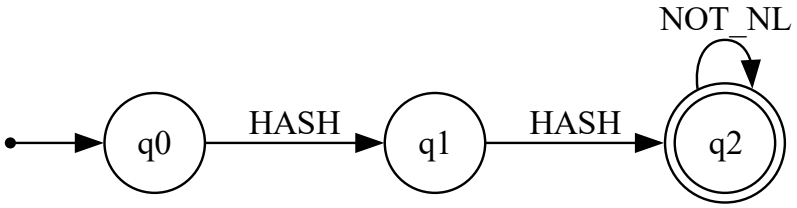
2.5 Integer



2.6 Punctuator

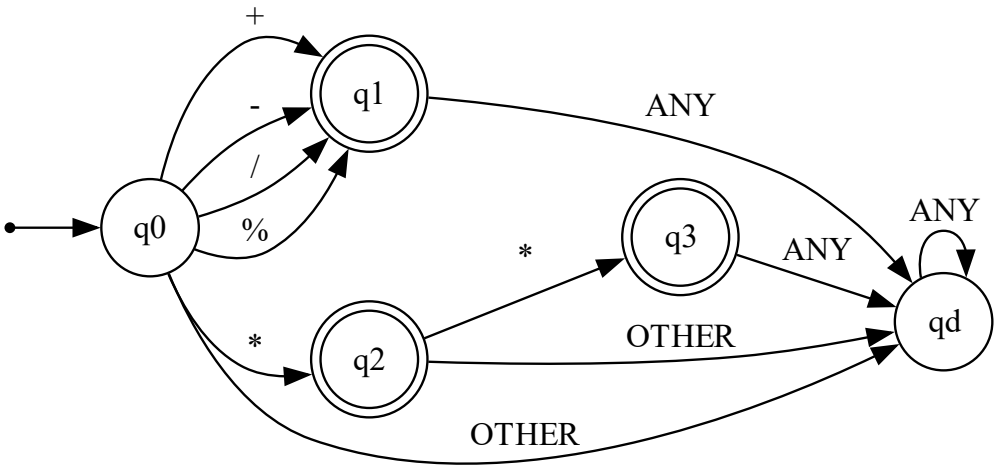


2.7 Single-line comment

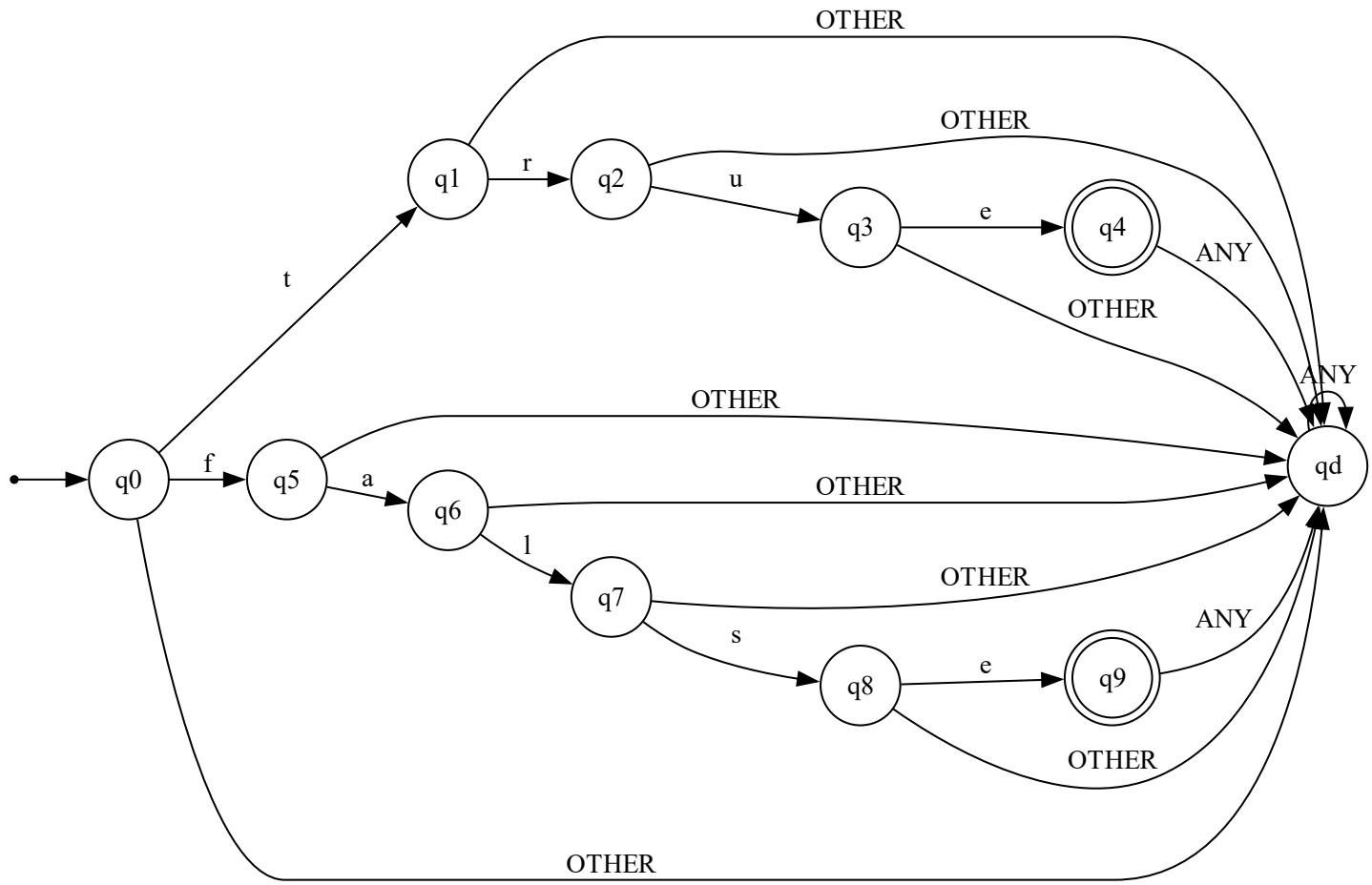


3. DFA

3.1 Arithmetic operators



3.2 Boolean



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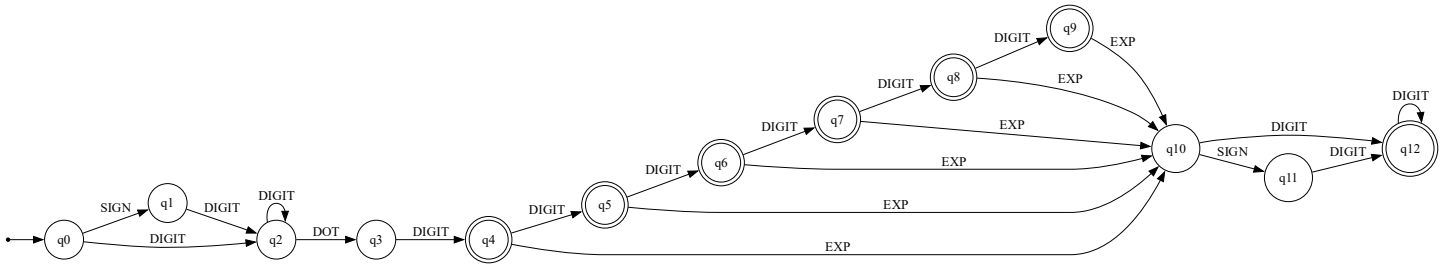
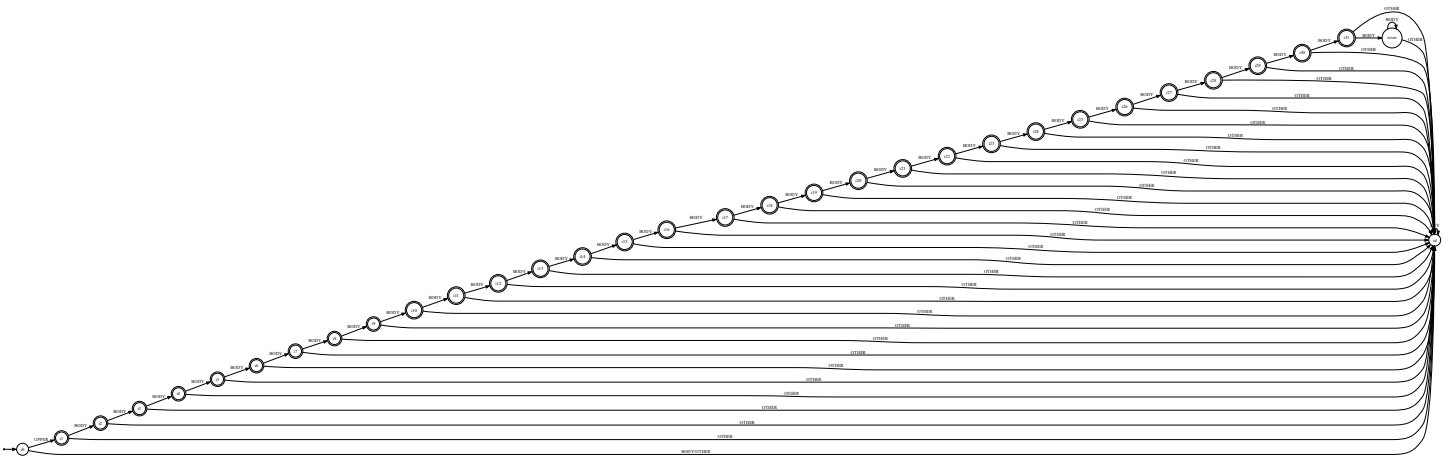
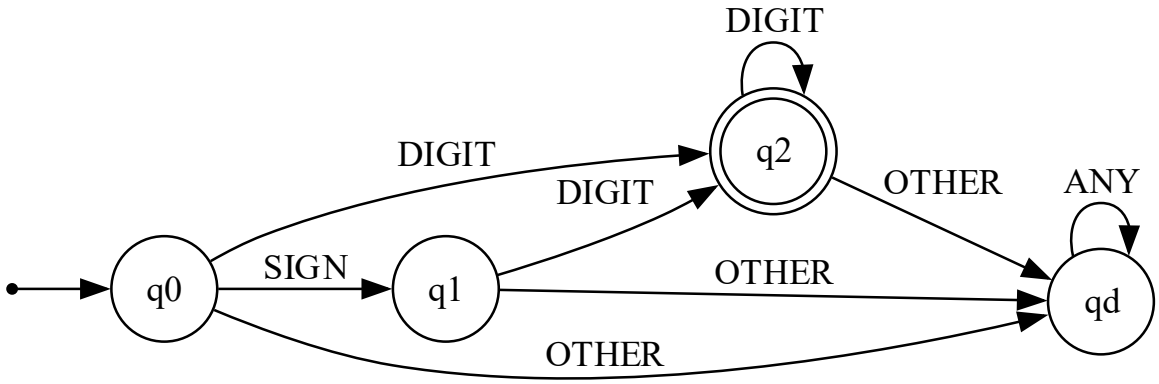


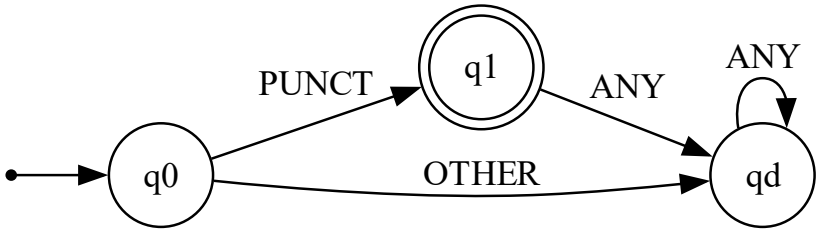
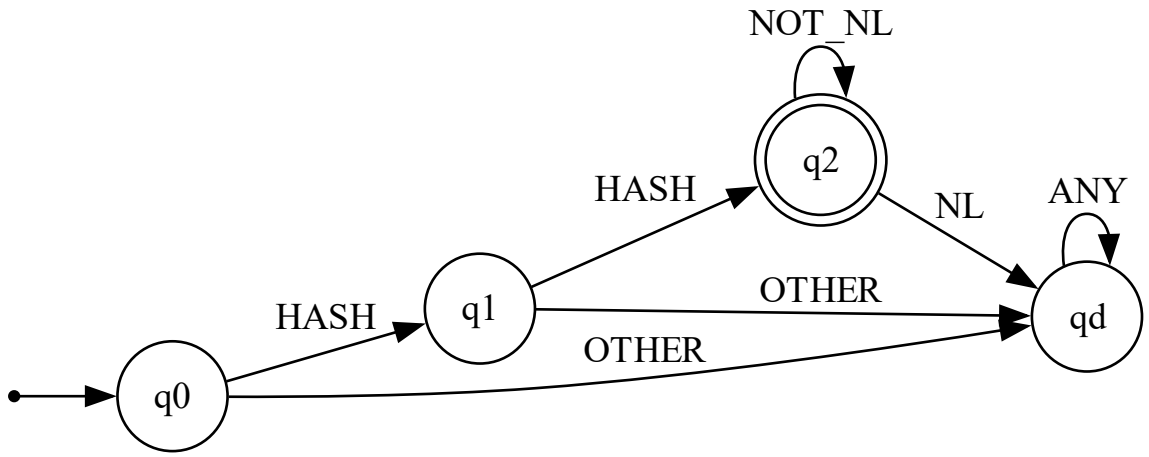
3.4 ID



3.5 Integer



3.6 Punctuator



3.7 Single-line comment